

LANCELIN, Australia

WMO: 956060

Lat: 31.016S

Lon: 115.330E

Elev: 5

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	8.1	0.4	3.9	17.5	2.1	4.4	15.7	14.6	15.9	13.5	16.3	2.6	90	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.7	33.9	20.4	31.5	20.3	29.6	20.3	24.2	28.7	23.3	27.6	22.5	26.8	5.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	17.6	27.0	21.9	16.6	26.0	21.0	15.7	25.2	73.3	29.0	69.3	27.5	66.3	26.8	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.0	10.2	9.3	DB	4.6	39.9	0.9	2.2	3.9	41.5	3.4	42.8	2.9	44.1	2.2	45.7
			WB	3.6	25.7	1.0	2.0	2.9	27.2	2.3	28.3	1.7	29.5	1.0	31.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	22.9	23.6	22.6	20.2	17.9	15.8	14.8	14.8	15.5	17.1	19.7	21.5
	DBStd	4.01	2.86	3.07	3.10	2.54	2.33	1.98	2.04	1.73	1.85	2.38	2.48	2.98
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	496	1	0	2	8	35	79	110	108	86	53	11	3
	CDD10.0	3231	401	379	391	306	246	173	149	151	166	220	292	358
	CDD18.3	685	143	147	135	64	22	2	1	1	2	14	53	102
	CDH23.3	4274	922	1124	929	294	57	2	1	1	7	76	256	606
	CDH26.7	1351	299	392	298	63	6	0	0	0	1	19	73	200
(14) Wind	WSAvg	4.7	5.3	5.0	4.7	4.0	4.0	4.4	4.6	4.4	4.8	4.8	5.1	5.4
(15) Precipitation	PrecAvg	600	3	11	14	31	90	127	125	93	57	30	21	6
	PrecMax	754	29	93	69	124	216	219	204	187	116	87	71	34
	PrecMin	443	0	0	0	1	16	38	47	18	13	4	1	0
	PrecStd	92	6	20	16	23	48	45	41	30	26	18	19	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	36.8	36.0	31.4	27.3	23.2	22.1	22.1	24.3	29.2	32.0	35.8
		MCWB	21.5	21.4	20.8	19.0	16.8	15.4	16.3	16.5	16.2	17.2	18.0	19.7
	2%	DB	32.8	33.2	31.9	28.4	24.8	21.6	20.1	20.5	21.6	25.1	28.6	31.2
		MCWB	20.7	21.1	20.2	18.4	17.6	15.7	15.5	15.9	16.3	17.6	18.1	19.5
	5%	DB	29.7	30.8	29.7	26.2	23.3	20.4	19.1	19.4	20.4	22.9	26.0	28.5
		MCWB	21.1	21.2	20.5	18.9	17.4	15.5	15.3	15.6	16.1	17.4	18.2	19.6
	10%	DB	27.6	28.8	27.7	24.8	22.1	19.5	18.3	18.5	19.4	21.4	24.0	26.3
		MCWB	21.3	21.4	20.4	18.9	17.2	15.3	15.0	15.0	15.7	16.8	18.4	19.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	26.0	25.0	23.1	20.7	18.6	18.4	18.7	18.7	20.7	21.6	23.3
		MCDB	30.1	30.5	29.2	25.6	23.3	20.7	20.0	20.6	21.8	24.2	25.5	28.0
	2%	WB	24.0	24.4	23.4	21.8	19.7	17.7	17.1	17.5	17.8	19.2	20.6	22.2
		MCDB	28.5	28.9	27.9	24.7	22.5	20.0	19.0	19.4	20.4	22.5	24.7	27.2
	5%	WB	22.9	23.4	22.5	20.9	18.9	16.9	16.3	16.6	16.9	18.3	19.8	21.4
		MCDB	27.0	28.0	26.9	24.1	21.9	19.4	18.4	18.6	19.6	21.6	23.9	26.1
	10%	WB	21.9	22.5	21.7	20.0	18.0	16.1	15.5	15.7	16.1	17.5	18.9	20.6
		MCDB	26.1	27.0	25.9	23.2	21.2	18.7	17.8	17.9	18.8	20.7	22.9	25.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	9.3	8.8	8.5	7.7	7.2	7.6	7.6	8.4	8.0	7.8
		MCDBR	13.4	12.7	12.8	11.4	10.3	9.4	8.3	9.5	10.2	12.0	13.1	12.7
		MCWBR	7.2	6.3	7.0	7.0	6.6	5.7	5.7	6.4	6.8	7.4	7.4	6.8
	5% WB	MCDBR	9.3	9.0	9.4	8.4	7.9	7.1	6.4	7.7	8.4	9.3	9.3	9.6
		MCWBR	6.2	5.5	6.5	6.9	6.2	5.2	5.4	6.3	6.4	6.6	6.1	5.9
(40) Clear-Sky Solar Irradiance	taub		0.347	0.346	0.332	0.321	0.310	0.301	0.301	0.309	0.328	0.328	0.330	0.335
	taud		2.557	2.578	2.603	2.624	2.620	2.630	2.620	2.591	2.517	2.540	2.556	2.560
	Ebn at Noon		993	975	956	917	879	864	880	916	944	982	1003	1007
	Edh at Noon		108	103	94	83	75	70	73	84	101	106	108	109
(44) All-Sky Solar Radiation	RadAvg		8.24	7.47	6.20	4.62	3.50	2.89	3.04	3.93	5.21	6.70	7.87	8.46
	RadStd		0.39	0.24	0.31	0.25	0.22	0.19	0.18	0.17	0.27	0.27	0.26	0.33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page